CASE STUDY 4: Student Performance Trend

Python Activity-Based Learning

1. Problem Statement

Write a Python program to compare a student's marks in two tests and identify whether the student's performance improved, declined or remained the same.

2. Objectives

• Read marks from two tests.

• Compare the marks.

• Display the performance trend.

3. Assumptions

• Both marks are between 0 and 100.

• The second test is compared with the first test.

4. Algorithm

1. Start.

2. Read first-test and second-test marks.

3. If second marks are greater, display Improved.

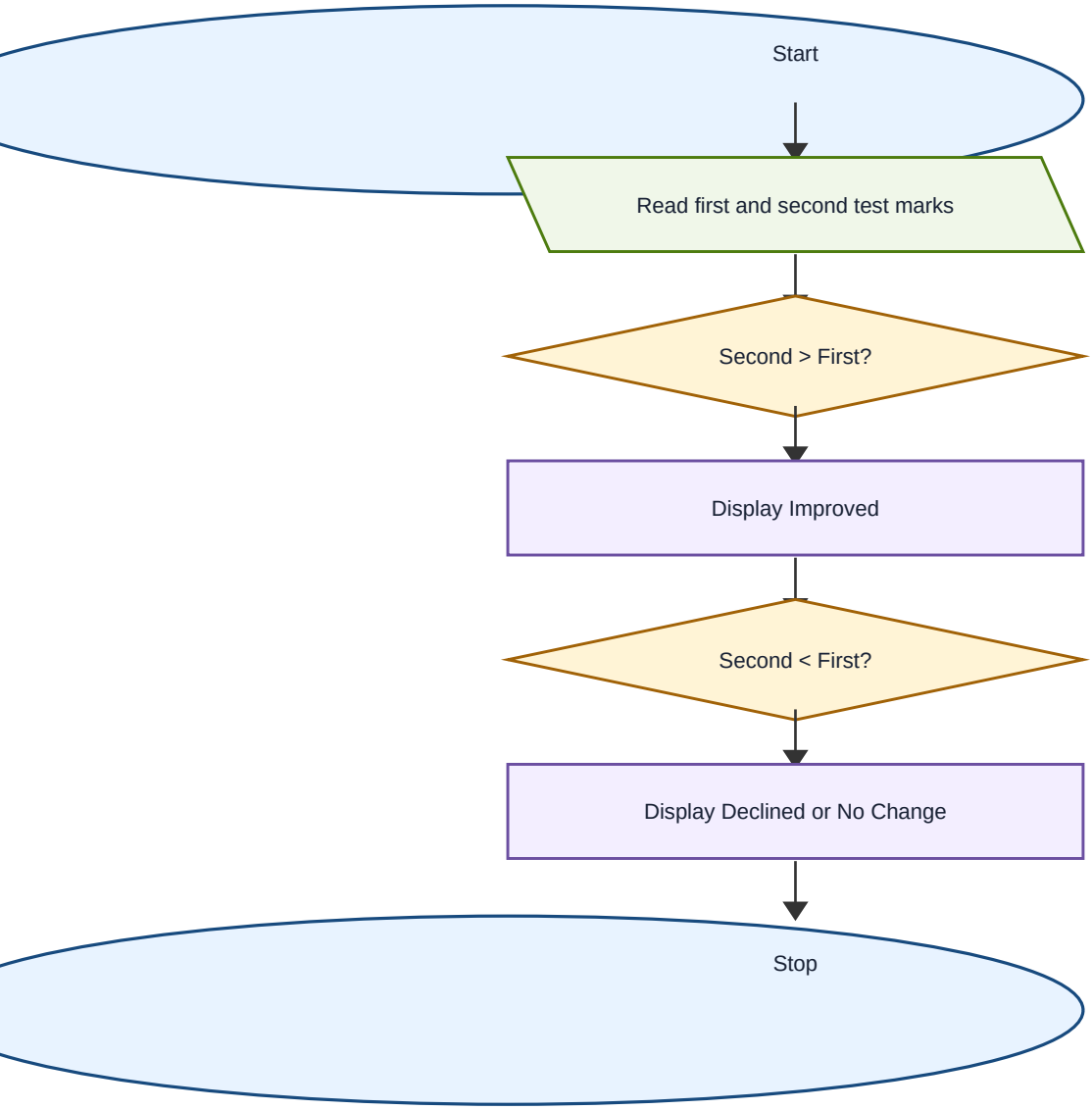
4. If second marks are lower, display Declined.

5. Otherwise, display No Change.

6. Stop.

5. Flowchart

Oval = Start/Stop; parallelogram = Input; rectangle = Process; diamond = Decision.



6. Pseudocode

```
BEGIN
  INPUT first, second
  IF second > first THEN
      PRINT "Performance Improved"
  ELSE IF second < first THEN
      PRINT "Performance Declined"
  ELSE
      PRINT "No Change"
  END IF
END
```

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7. Python Source Code

```
first = float(input("Enter first test marks: "))
second = float(input("Enter second test marks: "))

if first < 0 or first > 100 or second < 0 or second > 100:
    print("Enter marks between 0 and 100")
elif second > first:
    print("Performance Improved")
elif second < first:
    print("Performance Declined")
else:
    print("No Change")
```

8. Test Cases

Input	Expected Output
First 65; Second 80	Performance Improved
First 75; Second 60	Performance Declined
First 70; Second 70	No Change

9. Sample Output

Run the Python program and enter the values from one of the test cases above. The displayed result should match the expected output.

10. Conclusion

The program compares two values using relational operators and displays the student's performance trend.